

1. Boost converter is shown in Fig.2, consider all components to be ideal. Let input voltage  $V_d$  be 18V, output voltage  $V_o = 30V$ , and  $f_s = 20kHz$ .

Boost 变换器如图 2 所示, 考虑所有元件都是理想的。设输入电压  $V_d$  为 18V, 输出电压  $V_o$  为 30V, 开关频率为 20kHz。(8 分)

- (1) Derive the relationship between the output voltage  $V_o$  and input voltage  $V_d$ , and derive the expression of output voltage ripple ratio  $\Delta V_o/V_o$  when the inductor current is continuous. 推导电感电流连续时输出电压  $V_o$  与输入电压  $V_d$  之间的关系以及  $\Delta V_o/V_o$  的表达式。(2 分)
- (2) If the output voltage ripple ratio  $\Delta V_o/V_o$  is less than 5%, please calculate the minimum value of  $C$  in this converter when  $P_o = 180W$ . 如果输出电压纹波比  $\Delta V_o/V_o$  小于 5%, 请计算  $P_o = 180W$  时 Boost 变换器中电容  $C$  的最小值。(2 分)
- (3) Calculate  $L_{min}$  that will keep the converter operating in a continuous-conduction mode if  $P_o \geq 180W$ . 计算在  $P_o \geq 180W$  的情况下, 能够使变换器工作在 CCM 模式下的最小电感。(2 分)
- (4) With the  $L_{min}$  calculated in (3), please draw the waveforms of  $v_T$ ,  $i_L$  when  $P_o = 180W$ , then mark the value of the voltage stress of the switch T and the average inductor current  $I_L$ . 基于题(3)所计算出的  $L_{min}$ , 请画出  $P_o = 180W$  时  $v_T$ ,  $i_L$  波形, 并标出开关管 T 的电压应力和电感电流平均值  $I_L$  的大小。(2 分)

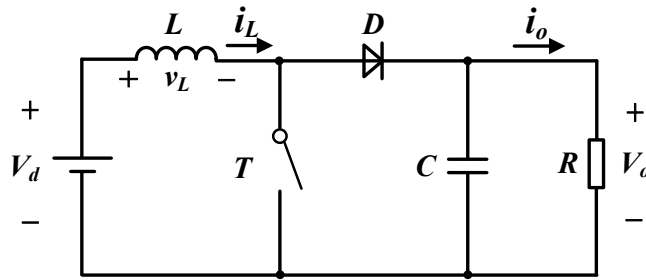


Fig.2 Boost converter

参考答案:

(1) 根据伏秒平衡推导:

$$V_d t_{on} + (V_d - V_o) t_{off} = 0$$

$$V_d D T_s + (V_d - V_o)(1 - D) T_s = 0$$

$$\frac{V_o}{V_d} = \frac{1}{1 - D}$$

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{I_o D T_s}{C} = \frac{V_o D T_s}{RC}$$

$$\frac{\Delta V_o}{V_o} = \frac{DT_s}{RC}$$

$$(2) \quad \frac{\Delta V_o}{V_o} \leq 5\% \quad \text{即} \quad \Delta V_o \leq 1.5V$$

$$D = 1 - \frac{V_d}{V_o} = 0.4, \quad I_o = P_o / V_o = 180 / 30 = 6A, \quad T_s = 50 \mu s$$

$$\Delta V_o = \frac{\Delta Q}{C} = \frac{I_o D T_s}{C}$$

$$\text{代入得} \quad C_{min} = 80 \mu F$$

$$(3) \quad \frac{V_o}{V_d} = \frac{1}{1-D}, \quad D = 1 - \frac{V_d}{V_o} = 0.4, \quad T_s = 1 / f_s = 50 \mu s$$

$$\text{最小输出电流} \quad I_{o,min} = P_o / V_o = 180 / 30 = 6A = I_{oB}$$

$$\text{由} \quad I_{oB} = \frac{T_s V_o}{2L} D(1-D)^2 \quad \text{代入占空比} \quad D,$$

$$\text{则} \quad L_{min} = 18 \mu H$$

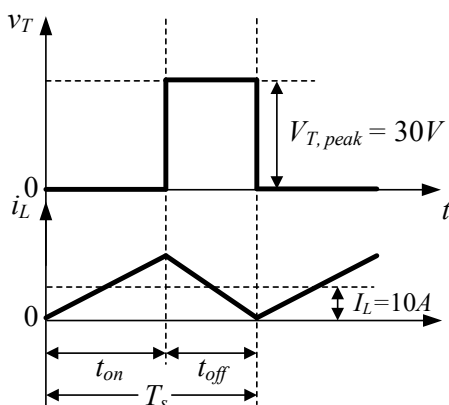
$$(4) \quad V_{T,peak} = 30V$$

根据题意，此时变换器工作在 BCM 模式下

$$L_{min} = 18 \mu H$$

$$I_{LB} = \frac{i_{L,peak}}{2} = \frac{1}{2} \frac{V_d}{L} t_{on} = \frac{T_s V_o}{2L} D(1-D) \quad \text{或} \quad I_{oB} = (1-D) I_{LB}$$

$$I_L = I_{LB} = 10A$$



2. In a Buck-Boost converter, consider all components to be ideal. Let input voltage  $V_d$  be 32 ~ 72V, output voltage  $V_o = 48V$ , and  $f_s = 50kHz$ ,  $L = 100\mu H$ .

在 Buck-Boost 变换器中，考虑所有元件都是理想的。设输入电压  $V_d$  为 32 ~ 72V，输出电压为 48V，开关频率为 50kHz，电感为 100 $\mu$ H。(9 分)

(1) Draw the Buck-Boost converter circuit diagram. 画出 Buck-Boost 变换器电路图。(1 分)

(2) Draw the waveforms of  $v_L$ ,  $i_L$ ,  $i_C$ . 画出  $v_L$ ,  $i_L$ ,  $i_C$  的波形。(3 分)

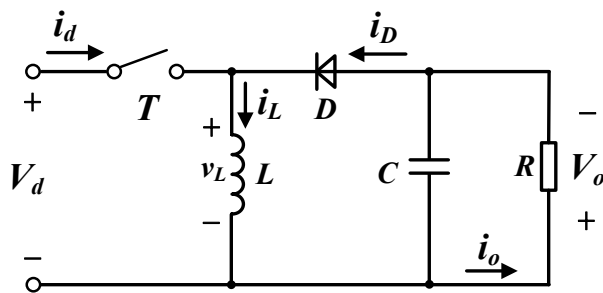
(3) Calculate the minimum output power  $P_{omin}$  that will keep the converter operating in a continuous-conduction mode for all range of  $V_d$ . 计算全输入电压范围内能够使变换器工作在 CCM 模式下的最小输出功率  $P_{omin}$ 。(2 分)

(4) What is the voltage stress of the switch. 开关管的电压应力是多少？(1 分)

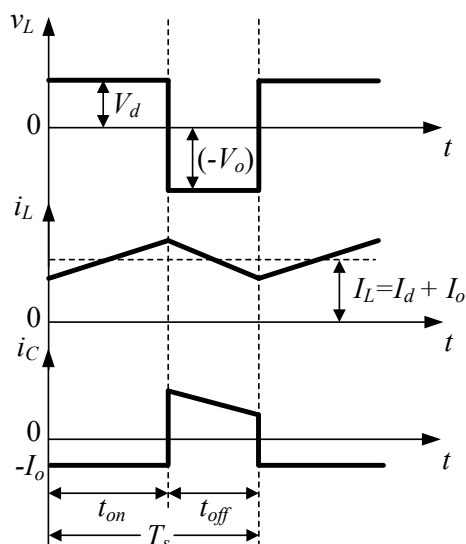
(5) What is the advantage and disadvantage of this converter. 上述变换器的优缺点是什么？(2 分)

参考答案：

(1) Buck-Boost 变换器



(2)  $v_L$ ,  $i_L$ ,  $i_C$  波形



$$(3) \quad \frac{V_o}{V_d} = \frac{D}{1-D}, \quad D = \frac{V_o}{V_d + V_o}, \quad T_s = 1/f_s = 20 \mu s$$

$$D_{\min} = 0.4, \quad D_{\max} = 0.6$$

$$\text{由 } I_{oB} = \frac{T_s V_o}{2L} (1-D)^2$$

最小的占空比  $D_{\min}$  对应最大的  $I_{oB\max}$ :

$$\text{得 } I_{oB\max} = 1.728 \text{ A}$$

$$P_{\min} = I_{oB\max} \times V_o = 82.944 \text{ W}$$

$$(4) \text{ 开关管电压应力: } V_o + V_{d,\max} = 120 \text{ V}$$

(5) 优点: 电路可升压可降压。缺点: 输入和输出反极性, 开关管的电压应力较大, 对开关管要求高。

**3. For the Flyback converter, input voltage is  $V_d$ , output voltage is  $V_o$ .**

对于一个反激变换器, 输入电压是  $V_d$ , 输出电压是  $V_o$ 。

**(1) Please draw the circuit diagram of flyback converter.**

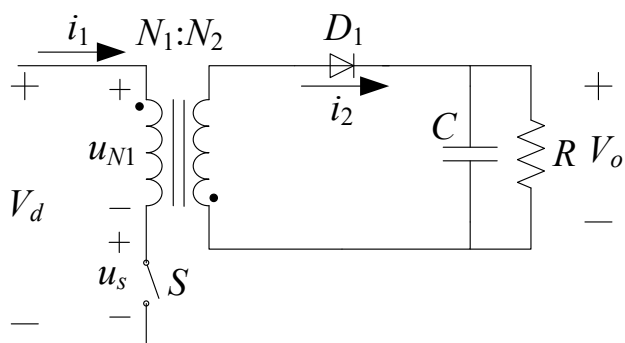
(1)请画出反激变换器电路图。(1 分)

**(2) Please derive the relationship between  $V_o$  and  $V_d$  under CCM operation mode.**

(2)请推导电感电流连续模式下输出电压  $V_o$  与输入电压  $V_d$  之间的关系。(2 分)

参考答案:

(1)



(1 分)

(2) 根据  $V = N \frac{d\Phi}{dt}$

故有  $d\Phi = \frac{1}{N} V dt$

在 S 导通期间，磁通变化量

$$\Delta\Phi = \frac{1}{N_1} V_d t_{on}$$

在 S 截止期间，磁通变化量

$$\Delta\Phi = \frac{1}{N_2} V_o t_{off}$$

在导通和截止期间。磁通的变化量应相等，故有

$$\frac{1}{N_1} V_d t_{on} = \frac{1}{N_2} V_o t_{off}, \quad (1 \text{ 分})$$

因此，  $\frac{V_o}{V_d} = \frac{N_2}{N_1} \frac{D}{1-D}$  , (1 分)

**4. A forward converter with a demagnetizing winding ( $N_3=N_1$ ) operates in a continuous-conduction mode,  $V_d = 45\sim 60V$ ,  $V_o=30V$ ,  $f_s=100kHz$ ,  $P_{load}$  is  $30W\sim 60W$ . Assume all components to be ideal except the transformer magnetization inductance. Please answer the following questions:**

一个正激变换器工作在电感电流连续模式，其磁通复位绕组匝数  $N_3=N_1$ ,  $V_d = 45\sim 60V$ ,  $V_o=30V$ ,  $f_s=100kHz$ ,  $P_{load}$  为  $30W\sim 60W$ 。假设变换器的所有元器件均为理想元件，除了变压器励磁电感。请回答以下问题：

**(1) Draw the circuit diagram of this converter.**

请画出正激变换器电路图。(1 分)

**(2) Please derive the relationship between  $V_o$  and  $V_d$ .**

请推导输出电压  $V_o$  与输入电压  $V_d$  之间的关系。(2 分)

(3) Calculate  $N_2/N_1$  if this turns ratio is desired to be as small as possible.

计算  $N_2/N_1$  的最小值。(2 分)

(4) Calculate the minimum value of the filter inductance  $L$ .

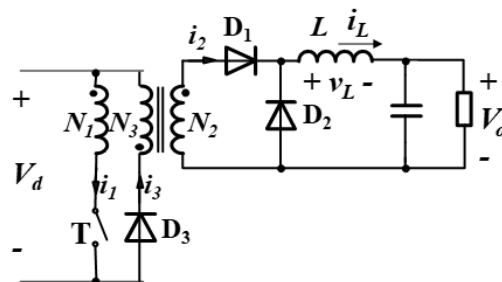
计算滤波电感  $L$  的最小值。(2 分)

(5) Calculate the peak voltage of switch  $T$ .

计算开关管  $T$  承受的峰值电压。(1 分)

参考答案:

(1)



(1 分)

(2)

$$\therefore \left( \frac{N_2}{N_1} V_d - V_o \right) T_{on} - V_o T_{off} = 0 \quad (2 \text{ 分})$$

$$\therefore \frac{V_o}{V_d} = \frac{N_2}{N_1} D$$

(3)

$$\therefore D_{\max} = \frac{1}{1 + \frac{N_3}{N_1}} = 0.5$$

$$\frac{V_o}{V_d} = \frac{N_2}{N_1} D \quad (2 \text{ 分})$$

$$\therefore \frac{N_1}{N_2} = D \frac{V_d}{V_o} \leq 0.5 \frac{V_{d_{\min}}}{V_o} = \frac{3}{4}$$

$$\therefore \frac{N_2}{N_1} \geq \frac{4}{3}$$

(4) 为了保持连续, 需要  $I_L \geq 0.5 \Delta i_L$ , 有

$$I_{o\min} = \frac{P_{o\min}}{V_o} = I_{L\min} \geq \frac{1}{2} \frac{\left(V_d \frac{N_2}{N_1} - V_o\right) D T_s}{L}$$

$$L \geq \frac{1}{2} \frac{\left(V_d \frac{N_2}{N_1} - V_o\right) D}{f_s I_{o\min}} = \frac{1}{2} \frac{V_o (1-D)}{f_s I_{o\min}} \quad (2 \text{ 分})$$

$$\therefore \frac{N_2}{N_1} = \frac{4}{3} \Rightarrow \frac{V_o}{V_{d\max}} \leq \frac{N_2}{N_1} D \leq \frac{V_o}{V_{d\min}} \Rightarrow \frac{3}{8} \leq D \leq \frac{1}{2}$$

$$\therefore L_{\min} = \frac{1}{2} \frac{V_o (1-D_{\min})}{f_s I_{o\min}} = 93.75 \mu H$$

(5)

$$V_T = \left(1 + \frac{N_1}{N_3}\right) V_d \quad (1 \text{ 分})$$

$$\therefore V_{T_{\max}} = 120V$$